



5782 - Abell 1689: Opening a Unique Window on Lensed Galaxies at Cosmic Noon

Cycle: 3, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Dr. Adelaide Marie Claeysens (PI) (ESA Member)	Stockholm University
Prof. Brian Siana (CoI) (CoPI) (US Admin CoI)	University of California - Riverside
Dr. Angela Adamo (CoI) (ESA Member)	Stockholm University
Dr. Miroslava Dessauges-Zavadsky (CoI) (ESA Member)	University of Geneva, Department of Astronomy
Dr. Johan Pierre Richard (CoI) (ESA Member)	Centre de Recherche Astrophysique de Lyon
Dr. Matteo Messa (CoI) (ESA Member)	INAF - Osservatorio di Astrofisica e Scienza dello Spazio
Dr. Anahita Alavi (CoI)	California Institute of Technology
Dr. Christopher Robert Snapp-Kolas (CoI)	Biola University
Ms. Thi Thai Tran (CoI) (ESA Member)	CNRS, Laboratoire d'Astrophysique de Marseille
Dr. Ilias Goovaerts (CoI)	Space Telescope Science Institute
Roser Pello (CoI) (ESA Member)	CNRS, Laboratoire d'Astrophysique de Marseille
Dr. Jorrit Matthee (CoI) (ESA Member)	Institute of Science and Technology Austria
Dr. Tuan-Anh Pham (CoI)	Vietnam National Space Center
Ye Lin (CoI)	University of California - Riverside
Dr. Marc Rafelski (CoI)	Space Telescope Science Institute

OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1		NIRCam Imaging	(1) ACO-1689

ABSTRACT

JWST Proposal 5782 (Created: Friday, May 2, 2025, 1:00:08PM Eastern Standard Time) - Overview

Galaxies at cosmic noon (CN, $z \sim 2$) are characterized by irregular morphologies dominated by compact and bright stellar clumps. The role of these clumps in galaxy morphological evolution and their contribution to the total galaxy mass buildup are so far poorly understood and widely debated. The combination of JWST/NIRCam high sensitivity and resolution with the strong gravitational lensing effect produced by galaxy clusters has opened a new window on the internal physical processes at play in distant galaxies. We propose deep NIRCam images of the massive lensing cluster Abell1689 to probe the physical properties of star-forming clumps and galaxies at CN, a key time for galaxy evolution. A1689 is far superior to the Hubble FF clusters to study CN galaxies as it is known to produce extremely high magnifications over a large volume of Universe at $1 < z < 4.5$. These observations will enable us to study > 1000 galaxies over a wide range of masses and resolve their clump and star cluster populations across CN down to 1 pc scale and 10^5 solar mass. By extending the wavelength coverage to the IR and combining broad-band and medium-band filters, we will be able to derive robust clump/cluster physical properties (size, age, mass, boundness). Clumps trace star-formation and play a major role in the stellar feedback at a galactic scale. We will answer questions related to diverse formation mechanisms and survival time scales of clumps as a function of galaxy mass. We expect to detect young and intermediate age proto-globular clusters in the many giant arcs produced by A1689, enabling us to constrain their redshift formation.

OBSERVING DESCRIPTION

The goal of this program is to obtain deep NIRCam broadband (F115W, F150W, F200W, F277W and F356W) and medium-band (F182M and F300M) imaging of the highly magnified regions behind the lensing cluster, Abell 1689. To obtain 5-sigma point source sensitivities (in apertures matching the PSF resolution) of ~ 29.7 (AB mag), all broadband filters have an exposure time of 7537s, except for F356W, which has twice the exposure time, 15,074s. The medium band filters have $\sim$ double the exposure time, 14,881s. This is to achieve similar point-source sensitivity (in f_{ν}) in all filters, and to achieve the highest surface brightness sensitivity to light from older stellar populations at Cosmic Noon ($1 < z < 4.5$).

Because the area of high lensing magnification is so large in the target cluster, we choose not to center the cluster in one module. Instead, we will image it in both modules and use a Fullbox 6 Tight dither pattern to cover the gap between modules. This will result in decreased (1/3) exposure in the very center of the cluster, but that area is lower priority as it has the large cluster galaxies and lower magnification.

For the filters with ~ 7.5 ks of total exposure, we're choosing to use 8 groups of the SHALLOW4 readout pattern. For the longer exposures in the medium band, we're using 4 groups, 2 integrations with the MEDIUM8 readout pattern. To better sample the PSF at the shorter wavelengths, we are using 3 sub-pixel positions at each dither pointing.

We use a special requirement to constrain the PA to 116. This position angle avoids significant contamination from "claws" and also aligns the

longer side of the mosaic with the area of enhanced magnification of Abell 1689 (which is also elongated NE-SW).

Proposal 5782 - Targets - Abell 1689: Opening a Unique Window on Lensed Galaxies at Cosmic Noon

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	ACO-1689	RA: 13 11 30.0000 (197.8750000d) Dec: -01 20 31.20 (-1.34200d) Equinox: J2000	Epoch of Position: 2000	
<i>Comments: This object was generated by the target selector and retrieved from the SIMBAD database.</i> <i>Category=Clusters of Galaxies</i> <i>Description=[Abell clusters]</i>					

Proposal 5782 - Observation 1 - Abell 1689: Opening a Unique Window on Lensed Galaxies at Cosmic Noon

Observation	Proposal 5782, Observation 1										Fri May 02 18:00:08 GMT 2025
	Diagnostic Status: Warning										
Observing Template: NIRCcam Imaging											
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 1:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 1:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 1:4) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 1:5) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 1:6) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 1:7) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 1:8) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	ACO-1689	RA: 13 11 30.0000 (197.8750000d)			Epoch of Position: 2000					
			Dec: -01 20 31.20 (-1.34200d)								
			Equinox: J2000								
Comments: This object was generated by the target selector and retrieved from the SIMBAD database.											
Category=Clusters of Galaxies											
Description=[Abell clusters]											
Template	Module			Subarray				Target Placement			
	ALL			FULL				Module Gap			
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	FULLBOX		6TIGHT		STANDARD				3	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID	
	1	F182M	F300M	MEDIUM8	4	2	36	18	14881.163		
	2	F200W	F356W	SHALLOW4	8	1	18	18	7537.213		
	3	F115W	F277W	SHALLOW4	8	1	18	18	7537.213		
	4	F150W	F356W	SHALLOW4	8	1	18	18	7537.213		
Special Requirements	Sequence Visits within 53.0 Days										
	Aperture PA Range 116 to 116 Degrees (V3 116.07457694 to 116.07457694)										
Visits Same PA											